

Leakage and Label Noise Inflate UAV Flight-Log Diagnosis: A Verifiable Real-Failure Benchmark for ArduPilot

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Abstract

Automated root-cause diagnosis of small-UAV crashes from onboard flight logs is often reported with high accuracy. We show, on real ArduPilot failure logs, that such figures are produced by evaluation practice rather than diagnostic skill. First, we audit forum-mined crash labels against the discussion threads they came from. Of 55 logs previously treated as labelled, only 24 have a cause stated in their thread, and those agree only weakly with the original labels (Cohen's $\kappa = 0.39$). The audit also found a simulated flight listed as a real crash, labels swapped between files, and a label derived from post-impact sensor noise. Second, we release a benchmark of 34 real logs from 32 incidents. Every label cites a verbatim forum post and can be re-verified automatically. Third, in a pre-registered evaluation, the same classifiers score 0.78–0.97 macro-F1 when windows of one flight are split at random, and 0.06–0.07 when incidents are kept intact, against a random-guess baseline of 0.08. No evaluated method, including a hybrid rule/ML engine with temporal root-cause arbitration, exceeds chance. Error analysis shows the rule engine labels the downstream symptoms of a failed motor (a power drop, compensating outputs) instead of the failure itself. We release code, labels and a one-command reproduction.

1 Introduction

Operators of ArduPilot-based vehicles [2] usually diagnose crashes by posting a DataFlash log to the community forum [1], where experienced users read the telemetry by hand. Automating this step is attractive, and several open tools claim to do it. Measuring whether they work is harder than it looks:

- Real crash logs are scarce.
- Labels are usually mined from forum text.
- A single flight yields many correlated windows.

These are the classic conditions for evaluation leakage [6] and benchmark label noise [8].

This paper is a self-audit of one such tool, an open-source hybrid rule/ML ArduPilot log diagnoser. Its published accuracy fell from *1.0* to *chance* as evaluation errors were removed. We make four contributions:

1. **A label audit** of forum-mined crash labels against their source threads (Section 4).
2. **A verifiable benchmark:** each label carries a permalink and a verbatim quote, checked automatically (Section 5).
3. **A pre-registered leakage study** showing a ten-fold inflation from window-level splitting (Section 6).
4. **An error analysis** explaining why rule-based diagnosis names symptoms rather than causes (Section 7).

2 Related work

UAV fault datasets such as ALFA [7] and RflyMAD [9] provide fault-injection or simulated flights with known fault times. BASiC [3] provides simulated ArduCopter failures. Such data are valuable, but a near-perfect score on scripted simulator faults does not transfer to organic field crashes. The tool audited here reported exactly such a score. Leakage through non-independent splits is a documented cause of irreproducible ML results across science [6]. Label errors in test sets can reorder method rankings [8]. We document both effects in a new domain.

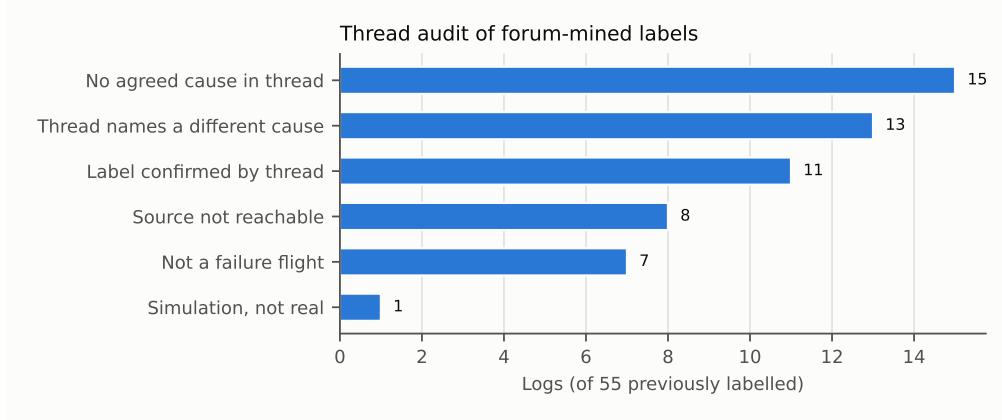


Figure 1: Outcome of re-labelling 55 previously labelled real logs from their source threads.

3 The audited system and its reporting history

The tool parses DataFlash logs and extracts 111 features in 5 s windows with 50% overlap. It combines a threshold rule engine, a tree classifier and an anomaly detector. A temporal arbitration step (CITA) chooses as root cause the diagnosis whose anomaly began earliest. Table 1 lists the headline figures the project published over time, and why each was withdrawn. All corrections are public in the repository.

Table 1: Reported headline accuracy of the audited tool over time.

Claim	Macro-F1	Why it does not hold
Simulated BASiC set	1.00	Scripted simulator faults, not real crashes
“Release” benchmark	0.81	All 6 evaluated logs were in the training set
Filename-grouped split	0.67	Windows of one incident crossed the split
URL-grouped split	0.56	Contradictory labels, mostly created by rule-engine relabelling
“Honest” candidate	0.50	Pool included simulated flights; artefacts never committed
This work (v3)	0.06–0.07	Real logs, incident-grouped, thread-verified labels

4 Label audit

The tool’s real-log labels came from forum threads. Some came from staff diagnoses, many from the search query used to find a thread, and three had been silently overwritten by the tool’s own rule engine. We fetched every cited thread and re-labelled each log from the thread alone.

Protocol. A label requires a post, by someone other than the original poster, that names a cause. Alternatively, the original poster may confirm a cause found after investigation. Competing causes without resolution give *undiagnosed*. Each label records:

- the post permalink, author and forum role;
- a verbatim quote;
- a certainty: *explicit* or *tentative* (hedged wording).

Label definitions follow the tool’s taxonomy. In particular, a sudden single-motor, ESC or propeller loss is `mechanical_failure`, not `motor_imbalance`.

Annotator disclosure. All labels were assigned by a large language model reading the threads. They have *not yet* been reviewed by a human; a 10-item human spot-check is prepared. A script re-verifies every quote against the thread text: 83 of 83 quotes matched the cached threads, and randomly re-fetched posts matched live.

Findings (Fig. 1):

- Of 55 logs, 24 have a cause stated in their thread: 11 confirm the old label and 13 change it (Cohen’s $\kappa = 0.39$ [4]).

Table 2: Benchmark v3 results (34 logs, 32 incidents). Model rows: mean $\pm$ s.d. over 5 seeds. Rule rows: 95% incident-cluster bootstrap CI.

Method	Random windows	Grouped by log	Grouped by incident
RandomForest	0.971 ± 0.000	0.088 ± 0.006	0.056 ± 0.009
ExtraTrees	0.971 ± 0.003	0.074 ± 0.002	0.069 ± 0.002
Logistic regression	0.777 ± 0.053	0.078 ± 0.007	0.069 ± 0.001
Macro-F1 (correct top-1 of 34)			
Rule engine	$0.062 [0.014, 0.139] (4)$		
Rules + fusion, CITA off	$0.054 [0.000, 0.124] (3)$		
Rules + fusion, CITA on	$0.000 [0.000, 0.000] (0)$		
Majority class	0.038		
Frequency-random (mean / 95th pct)	$0.076 / 0.152$		

- 15 threads never agree on a cause, 7 logs are not failure flights, 1 is a Gazebo simulation, and 8 sources were unreachable.
- The most common correction is `motor_imbalance` to `mechanical_failure`.
- In one thread, labels were swapped between the crash log and an uneventful post-crash flight.
- In another, a GPS label came from errors that a reviewer notes “occurred after it hit the ground”.

5 Benchmark

The benchmark covers in-flight failures, most of which ended in a crash; a few are failsafe or loss-of-thrust events that ended in a landing.

v3 combines the 22 audited logs that are usable (confirmed or relabelled, locally available, no within-incident conflict) with 12 previously unlabelled logs. Those 12 were labelled with the same protocol from 36 downloadable, hash-verified pool logs. The result is **34 logs from 32 incidents in 12 classes**, of which 14 labels are explicit. `mechanical_failure` is the largest class (10 logs).

Every log is identified by its SHA256 hash and source thread. Because redistribution rights for forum logs are unverified, we release links, hashes, derived features and labels, not raw logs, with a download script (in the spirit of [5]).

6 Pre-registered evaluation

The protocol and hypotheses were committed before any v3 result was computed. v1/v2 results were known beforehand, so the test is partially confirmatory.

Metric. Log-level macro-F1 over the classes present, using the maximum window probability per log. Uncertainty is a 95% incident-cluster bootstrap.

Splits. Three protocols differ only in grouping, each 5-fold repeated over 5 seeds:

- random windows;
- grouped by log file;
- grouped by incident.

Models and baselines.

- RandomForest, ExtraTrees and logistic regression (fixed hyper-parameters);
- the rule engine;
- rules with fusion, with CITA on and off;
- majority-class and frequency-random baselines.

Results (Fig. 2, Table 2):

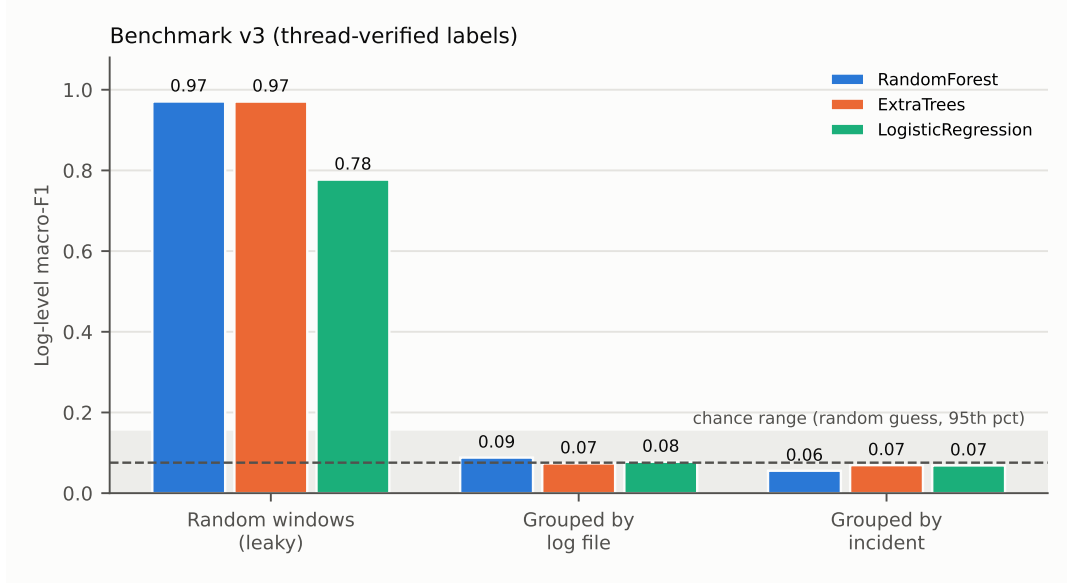


Figure 2: Same models, same data, three split protocols. The shaded band is the frequency-random baseline up to its 95th percentile.

- **H1 (leakage) is supported for every model.** Random window splits give 0.78–0.97; incident grouping gives 0.06–0.07.
- **H2 (no skill) is supported.** No method exceeds the 95th percentile of the frequency-random baseline (0.152).
- **H3 is supported.** CITA does not help: it changed 19 of 34 top-1 rule diagnoses, and correct answers fell from 3 to 0.
- **The explicit-label subset** (14 logs) gives the same picture: 2/14 correct for rules, 0/14 with CITA.
- **Earlier label versions** give the same staircase: $0.89 \rightarrow 0.14\text{--}0.17 \rightarrow 0.08\text{--}0.09$ on v1, and $0.99\text{--}1.00 \rightarrow 0.05$ on v2.

7 Error analysis

For the 10 verified `mechanical_failure` logs, where a motor, ESC or cable suddenly stopped producing thrust:

- **Rule engine:** predicts `power_instability` 5 times and `motor_imbalance` 2 times, and is correct once.
- **With CITA:** predicts `motor_imbalance` 6 times.

Both are *consequences* of the failure. Current drops when a motor stops, and the controller drives the remaining motors apart to compensate. Onset-ordering cannot separate them from the cause, because they begin within milliseconds of it. Better diagnosis therefore needs features that model the failed actuator directly: a commanded output saturating at maximum while its thrust contribution and current fall. Earliest-anomaly heuristics are not enough.

8 Limitations

- The benchmark is small (34 logs), class-imbalanced and multicopter-heavy, and it inherits forum selection bias.
- Labels come from a single automated annotator and have not been human-reviewed. Each is independently checkable, and a human spot-check is pending.
- Many causes are tentative in their threads.

- Fault onset times are not yet annotated.
- A fully confirmatory test on incidents posted after registration is future work.

9 Availability

Code, registry, thread annotations with quotes, derived features, result files and the pre-registration are public in the project repository. `python training/paper_eval.py` regenerates every table. Result files record the exact commit and input hashes.

References

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